

LEC 5 - The Local Sky

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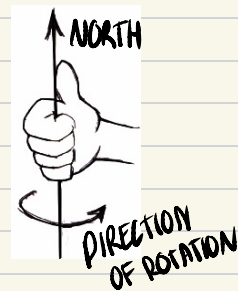
The right hand rule:

A technique to determine the direction of rotation of a planet. Your thumb points North. The direction of the curl of your fingers is the direction of rotation.

* We can use latitude (N/S) and longitude (E/W) to describe locations on different bodies (such as Earth)

- Planets orbit stars
- Stars orbit galaxy centers
- Galaxies move relative to each other
- Galaxies move as the universe expands

- Earth completes one rotation every 24 hours
- You can use the right hand rule to determine the direction of rotation



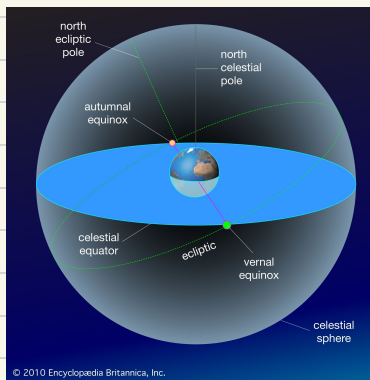
Coordinates on Earth

Latitude measures distance north or south of the equator.

Longitude measures east or west of an arbitrary chosen reference location.

The Celestial Sphere

Celestial poles and equator are projections of the earth's poles and equator into the sky.



How do you differentiate between North and South celestial pole?
You will see the respective pole over the horizon on the respective hemisphere (equator sees both poles)

Zenith - Straight Up (Altitude 90°)

Horizon - Where Sky meets ground (Altitude 0°)

Star Trails from long-exposure Photography

- Opening camera shutter for minutes or hours at a time causes all stars in photograph to turn into streaks portraying their movements across the photo time



Question: How would a star trail photo change between Toronto & the North Pole

North Celestial Pole

The center of rotation would change from directly above (North pole) to approximately a 44° angle from North

How can we describe the location of objects in the sky?

Altitude - Angular height of object above ground

Azimuth - Angular direction of object East of North

$\Rightarrow N = 0^\circ$ Azimuth

$\Rightarrow E = 90^\circ$ Azimuth

$\Rightarrow S = 180^\circ$ Azimuth

$\Rightarrow W = 270^\circ$ Azimuth

- We can use latitude (N/S) and longitude (E/W) to describe the locations of objects on different bodies (such as Earth)
- We can use altitude (vertical angle) and azimuth (horizontal angle) to describe the location of objects in the sky
- Long exposure photography turns stars from points to streaks representing their motion by opening the camera shutter for an extended period of time (open shutter lets light in)